

various parameters and an opto-interrupter sensor to count motor rotations and provide feedback.

The author's prototype, including motor-sensor arrangement, is shown in Fig. 1.

Circuit and working

Circuit diagram of the auto-reversible DC motor control is shown in Fig. 2. It is built around opto-interrupter module MOC7811 (IC1), ATmega16 microcontroller (IC2), 16 × 2 LCD (LCD1), motor driver L293D (IC3), 12V DC motor (M1) and a few other components.

Four push-to-on button/tactile switches (S1 through S4) are connected to port C pins PC0, PC1, PC6 and PC7 of IC2, so that when a button is pressed, it gives a high logic (1) input to the corresponding pin. These switches are used for various settings to control the motor, including up, down, enter and rotation.

LCD1 is connected to port A and port D of IC2. Its data pins are connected to port A, and the control pins EN and RS are connected to PD0 and PD1, respectively. LCD1 displays the number of rotations and speed of the DC motor.

Port D pin PD7 and port B pin PB3 of IC2 drive the DC motor through motor driver chip L293D. These pins are connected to input pins 2 and 7 of the L293D, and output pins 3 and 6 of the L293D are connected to the motor (M1) terminals.

PB3 and PD7 are PWM output pins. PWM output on these pins will vary the speed of motor. Also, when PB3 generates PWM, PD7 is off and the motor rotates in forward (clockwise) direction, and when PD7 generates PWM, PB3 is off and motor rotates in reverse (anticlockwise) direction.

Opto-interrupter sensor MOC7811 consists of an IR LED and a phototransistor. The internal IR LED is forward-biased to keep it continuously on. A current-limiting resistor (R1) of 330-ohm is used to limit the current flowing through it. Output of the phototransistor is given as input to the base of transistor T1 that is connected in switch mode. Final output of the sensor circuit is taken from collector of T1. This output is given to external interrupt 0 input pin PD2 of ATmega16.

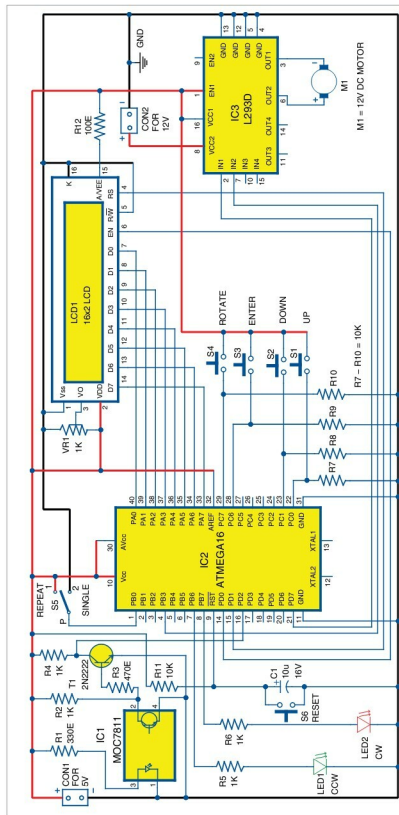


Fig. 2: Circuit diagram of the auto-reversible DC motor control